

Lecture Notes on Differential Equations and Transform Techniques

Lecture Notes for Spring 2025

The purpose of these lecture notes is to provide a concise summary of the key concepts covered in the course. They are designed to serve as a **quick reference** and should be used in conjunction with the core textbooks listed below. While some sections are adapted from these referenced texts, the notes are not intended to replace the original works, which offer more comprehensive explanations. To gain a deeper understanding, students are encouraged to refer to the textbooks listed.

References:

1. W. E. Boyce, R. C. DiPrima, D. B. Meade, *Elementary Differential Equations and Boundary Value Problems*, 9th edition, Wiley, 1997.
2. Brad G. Osgood, *Lectures on Fourier Transform and Its Applications*, IAS/Park City Mathematics Series / American Mathematical Society (AMS), 2019.
3. Charles Henry Edwards, David E. Penney, *Differential Equations and Boundary Value Problems: Computing and Modeling*, 4th edition, Pearson, 2000.
4. George F. Simmons, *Differential Equations with Applications and Historical Notes*, 2nd edition, CRC Press, 2016.
5. Ruel V. Churchill, *Fourier Series and Boundary Value Problems*, McGraw-Hill, 1941.
6. Gustav Doetsch, *Introduction to the Theory and Application of the Laplace Transformation*, Springer, 2012.
7. Morris Tenenbaum, Harry Pollard, *Ordinary Differential Equations*, Dover Books on Mathematics, Dover Publications, 1985.

EE1060: Differential Equations and Transform Techniques**Lecture Notes - 09****Topics covered :**

1. Introduction to Numerical Stability
2. Conditions for stability

- Numerical methods for solving an ODE of the form (1) approximate the solution at discrete points in the domain.

$$\frac{dy}{dx} = f(x, y), \quad y(x_0) = y_0 \quad (1)$$

The error at each point depends on several factors, with the [step size](#) being one of the most significant. [Numerical stability](#) refers to how errors propagate as the computations progress. More specifically, a numerical method is considered [stable](#) (for given conditions, such as the step size) if the errors do not grow exponentially or go unbounded as the calculation proceeds.

Typically, the stability of a numerical method is assessed using [linear stability analysis](#), which involves analyzing how errors propagate as iterations progress. The ODE used for linear stability analysis is

$$\frac{dy}{dx} = \lambda y \text{ and } y(0) = y_0 \quad (2)$$

where λ is a constant. Traditionally, for assessing stability, $\lambda < 0$ (in which case, the solution $y = y_0 e^{\lambda x}$ remains bounded) is considered for analysis. More importantly, a large negative value of λ indicates a scenario where the solution decays much faster (as shown in Fig. 1). This is one of the key characteristics of [stiff](#) differential equations. Thus, an ODE of the form (2) with a large negative value of λ is ideal for assessing the stability of a numerical

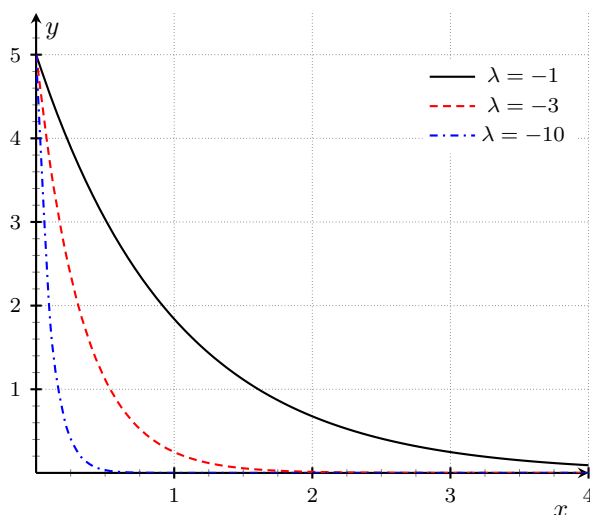


Fig. 1: Graph of analytical solution of ODE (2) for various λ with $y_0 = 5$

method. More specifically, it helps to analyze whether a numerical method can successfully handle fast-decaying solutions for various step lengths.

- A numerical method is **conditionally stable** if it is stable (i.e., errors do not grow unbounded) only for certain values of h . On the other hand, a numerical method is considered **A-stable** if it is stable for all values of h .
- **Linear stability assessment of Forward Euler method:** The update equation for the forward Euler method is

$$y_{n+1} = y_n + hf(x_n, y_n) \quad (3)$$

Application of the update equation to the test ODE (2), we get

$$y_{n+1} = y_n + h(\lambda y_n) \implies y_{n+1} = (1 + h\lambda) y_n \quad (4)$$

Equation (4) can be rewritten in terms of the initial condition y_0 as

$$y_{n+1} = (1 + h\lambda)^{n+1} y_0 \quad (5)$$

The factor $1 + h\lambda$ is often referred to as **Growth factor**. Observation into (5) indicates that if $|1 + h\lambda| > 1$, the solution grows unbounded (and so does the error). Thus the forward Euler method is stable if

$$|1 + h\lambda| < 1^1 \quad (6)$$

- **Linear stability assessment of Backward Euler method:** The update equation for the backward Euler method is

$$y_{n+1} = y_n + hf(x_{n+1}, y_{n+1}) \quad (7)$$

Application of the update equation to the test ODE (2) results in

$$y_{n+1} = y_n + h(\lambda y_{n+1}) \implies y_{n+1} = \frac{1}{(1 - h\lambda)} y_n \quad (8)$$

Rewriting (8) in terms of the initial condition gives

$$y_{n+1} = \left(\frac{1}{1 - h\lambda} \right)^{n+1} y_0 \quad (9)$$

For $\lambda < 0$, the growth factor $\frac{1}{1 - h\lambda}$ is always < 1 ensuring that the solution decays over time and errors do not grow unbounded. This indicates that the Backward Euler method is stable for all values of the step size h , and, as a result, it is classified as A-stable.

- **Linear stability assessment of RK2 method:** The update equations for the RK2 method is

$$y_{n+1} = y_n + \frac{1}{2}k_1 + \frac{1}{2}k_2 \text{ where } k_1 = hf(x_n, y_n) \text{ and } k_2 = hf(x_n + h, y_n + k_1) \quad (10)$$

¹In the practice problem set 3, you will need to derive the condition on h for which the method is stable

Application of update equation to ODE 2 gives

$$y_{n+1} = y_n + \frac{h}{2}\lambda y_n + \frac{h\lambda}{2}(y_n + hy_n) \implies y_{n+1} = \left(1 + h\lambda + \frac{h^2\lambda^2}{2}\right)y_n \quad (11)$$

Rewriting (11) in terms of the initial condition y_0 yields

$$y_{n+1} = \left(1 + h\lambda + \frac{h^2\lambda^2}{2}\right)^{n+1} y_0 \quad (12)$$

For $\lambda < 0$, the growth factor is $\left(1 + h\lambda + \frac{h^2\lambda^2}{2}\right)$ and as a result, RK2 method is stable if h is such that $\left|1 + h\lambda + \frac{h^2\lambda^2}{2}\right| < 1$. Thus, the RK2 method is conditionally stable.